(48)

(or) $100\% \xrightarrow[R=10\%]{SI=100\%} 200\% \Rightarrow t = 10$ yrs.

Ex:-3:- At what rate % per annum, a sum of money becomes double in 5 years on SI ?

Ans:- $A_1 = P + SI = 2P \Rightarrow SI = P$

$$SI = P \Rightarrow \frac{P \times 5 \times R}{100} = P$$

$$\Rightarrow R = \frac{100}{5} = 20\% \quad \text{(or)}$$

$$P = 100\% \Rightarrow 100\% \xrightarrow[5\text{ yrs}]{+100\%} 200\% \quad (R = 20\%)$$

Ex:-4:- $P = 25000$, $R = 10\%$, $n = 3$ years. on CI

(i) CI after 1 year (or) 1st year CI

(ii) 2nd year CI & CI after 2 years

(iii) 3rd year CI & CI after 3 years

(iv) A_2 after 3 years, 2 yrs & 1 yr ?

Ans:- $R = 10\% \Rightarrow$ factor multiplied is 1.1

(i) $\Rightarrow A_{2-1} = \left(1 + \frac{10}{100}\right)^1 \times P = 1.1 \times 25000 = 27500$

CI after 1yr / 1st yr CI $= A_{2-1} - P$

$= 27500 - 25000 = 2500/-$

(ii) CI after 2 yrs $= A_{2-2} - P$

$= P\left(1 + \frac{r}{100}\right)^2 - P = 25000(1.1)^2 - 25000$

$= \frac{100000}{4} \times 1.21 = \frac{1000 \times 121}{4} = 30250 - 25000 = 5250/-$

2nd year CI = 2 yrs CI − 1st yr CI

(49)

$= 5250 - 2500 = 2750$

(ii) 3 years CI $= A_{2-3} - P = P\left(1+\frac{R}{100}\right)^3 - P$

$= 25000\left(1+\frac{10}{100}\right)^3 - 25000 = \left[(1.1)^3 - 1\right] 25000$

$= 25000\left[1.331 - 1\right] = 25000 \times 0.331$

$= \frac{100000}{4} \times 0.331 = \frac{100 \times 331}{4} = 82.75 = 8275$

3rd year CI = 3 years CI − 2 years CI

$= 8275 - 5250 = 3025/-$

(iv) A_2 - after 1 year $= A_{2-1} = 27500$

A_2 " 2 " $= A_{2-2} = 30250$

A_2 " " " $= A_{2-3} = 25k(1.331) = 33275/-$

(or)

By using definition:-

$A_{2-1} = 25000 + 10\%$ of $25000 = 27500$

2500 → 1st year CI

1st yr CI + 10% of 1st yr CI = 2nd yr CI = 2500 + 10% of 2500

$= 2750$

3rd yr CI = 2nd year CI + 10% of 2nd yr CI

$= 2750 + 10\%$ of $2750 = 2750 + 275 = 3025/-$

$A_{2-1} = 27500$, $A_{2-2} = 27500 + 10\%$ of A_{2-1} (or) 2nd year CI

$= 27500 + 2750 = 30250$

$A_{2-3} = A_{2-2} + 10\%$ of A_{2-2} (or) 3rd yr CI $= 30250 + 3025 = 33275/-$

(or)

$\Big[\big[25000 + 10\% \text{ of } 25000\big] + 10\% \text{ of } 27500\Big] + 10\% \text{ of } \big(2750 + 2750\big) = 33275$

27500 → A_{2-1}; 2500 → 1st yr CI; 2750 → 2nd yr CI; 30251; 3025 → 3rd CI; A_{2-2}; A_{2-3}

(50)

Note:- In SI, there is no change in interest if interest is calculated annually (or) half yearly (or) quarterly (or) monthly (or) .... for a particular time period.

Note:- In CI, there is change in interest if calculated yearly, half yearly, quarterly .... for a period of time.

(i) Annually :- $A_2 = P\left(1+\frac{R}{100}\right)^n$

(ii) Half yearly :- n becomes 2n [because in a year 2 half years]

R becomes $\frac{R}{2}$ [Because interest becomes half per half-year]

$\Rightarrow A_2 = P\left(1+\frac{R/2}{100}\right)^{2n}$

(iii) Quarterly :- $n \to 4n$ & $R \to R/4$.

$\Rightarrow A_2 = P\left(1+\frac{R/4}{100}\right)^{4n}$

(iv) Monthly :- $n \to 12n$. & $R \to R/12$

$A_2 = P\left(1+\frac{R/12}{100}\right)^{12n}$

(v) Daily :- 1) L.Y 2) N.L.Y

1) Leap Year :- $n \to 366n$, $R \to R/366$

$A_2 = P\left(1+\frac{R/366}{100}\right)^{366N}$

2) Non-Leap Year :- $n \to 365$, $R \to R/365$

$A_2 = P\left(1+\frac{R/365}{100}\right)^{365n}$